Grape Diseases

Combined disease-wise reference

Eleven major diseases of grape (Vitis vinifera), India focus (Maharashtra, Karnataka, Andhra Pradesh, Tamil Nadu). Identification, causes, favourable conditions, management and verified remedies.

Compiled 19 October 2026. Primary sources: ICAR-National Research Centre for Grapes (CIB&RC label-claim list, revised 29/10/2024; Technical Bulletin No. 8); National Horticulture Board; NIPHM-DPPQS; university extension material. Chapter 11 (Esca) was added on 21 October 2026 from UC IPM and other international sources.

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About this document

Verification tags used in the chemical tables

V1 - Verified. The product appears on the ICAR-NRC Grapes list of chemicals with a CIB&RC label claim for grapes (A5, revised 29 October 2024) for this disease. Dose and pre-harvest interval (PHI) are reproduced as printed there; the label waiting period from the CIB&RC list as reproduced in DP22 (2021 data) is shown beside it.

V2 - ICAR / Government of India publication only (TB8, NHB or DP22). The product or practice is not on the A5 label-claim list for this disease, so confirm the current CIB&RC registration and label before use.

V3 - Registered on grape for this disease in the CIB&RC list as reproduced in DP22 (2021 data) but NOT on the NRCG export (EU-MRL) list A5. Only the label waiting period is available here, so the dose is 'per label'. Check the current label, and check residue limits before using in an export vineyard.

-- - No chemical option verified for this disease; none is recommended.

Important cautions

- The V1 doses and PHIs come from ICAR-NRC Grapes' list of chemicals with CIB&RC label claim for grapes (A5, revised 29 October 2024), which is advisory. The label waiting periods shown beside them come from the CIB&RC list as reproduced in the DPPQS IPM package (DP22; data of 01/01/2021). A5 PHIs are set with export residue limits in mind and are often longer than label waiting periods; if you export, use the stricter value. Check the live CIB&RC register (cibrc.gov.in) and the product label for the current registration, dose and waiting period.
- V3 products are on the CIB&RC list for grape in DP22 but not on the NRCG export list. Their doses were not available in the sources reviewed, so none is given; use the label, and check importing-country residue limits before using them in an export vineyard.
- Doses are for high-volume sprayers at about 1000 L/ha. The active ingredient per hectare should be kept as recommended even if spray volume changes (A5).
- Resistance has been detected in India against QoI and CAA fungicides in downy mildew and against triazoles in powdery mildew (A5). Products with these groups are marked with * and should be minimised, rotated and avoided in high-risk periods.
- The NRCG Technical Bulletin 8 (TB8, 2007) and the NHB note are older and are used for biology, timing and cultural practice. Where their product names or doses differ from A5 or DP22 (for example benomyl, Bayleton, Calaxin, dinocap, or copper at 3 g/L), the newer lists win and the older products are not reproduced. Biological products (Trichoderma, Bacillus) come from DP22 and TB8 and were not verified against the CIB&RC bio-pesticide register.
- Chapters 8-10 (black rot, bitter rot, dead arm) rest mainly on the National Horticulture Board note and the NIPHM pest list, which are older and give no verified chemical for these diseases.
- Use PPE, follow the label, rotate fungicide groups and keep products away from water bodies.
- No MPKV Rahuri, PDKV Akola, TNAU, FAO or KVK document specific to grape diseases was retrieved and used; the ICAR-NRCG (Pune) and DPPQS material is the primary Indian source.

Summary table

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Downy Mildew	Plasmopara viticola (Oomycete (fungus-like), rain-splash spread)	Sanitation at pruning; time protectant sprays to growth stage; rotate groups; limit systemic sprays; keep adjacent block pruning within 9 days	V1 options (33 on A5); 10 more V3
Powdery Mildew	Uncinula necator (Fungal (obligate parasite))	Open canopy; inner-canopy coverage; potassium nutrition; rotate systemic and non-systemic sprays; no sulphur at high temperature	V1 options (22 on A5); V2 potassium bicarbonate, oil; 6 more V3
Anthracnose (Bird's-eye Rot)	Elsinoe ampelina (Fungal (cane- and debris-borne; spread by rain and dew))	Remove and compost pruned wood; protect tender shoots; acidify carbendazim spray water	V1 options (11 on A5); 7 more V3

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Bacterial Leaf Spot	Xanthomonas campestris pv. viticola (Bacterial (spread in warm, humid weather))	Healthy propagative material; collect and burn infected parts; mancozeb/copper as for downy mildew	V1: kasugamycin + copper oxychloride; V2 copper, mancozeb
Alternaria Leaf Blight and Bunch Necrosis	Alternaria alternata (also A. tenuissima) (Fungal (weak pathogen; often follows heavy systemic fungicide use))	Limit systemic sprays; Trichoderma in humid September-October; leaf sanitation	None on A5; V2 Trichoderma only
Post-harvest Berry Rots (Botrytis, Aspergillus, Rhizopus and others)	Botrytis cinerea, Aspergillus niger, Rhizopus sp., Cladosporium, Alternaria, Botryodiplodia (Fungal (wound-entry rots in transit and storage))	Careful handling; precooling and cold chain; SO2 pads; Trichoderma before harvest	None on A5; V2 iprodione, Trichoderma
Leaf Rust	Phakopsora vitis (Fungal (rust; wind-borne urediniospores))	Scout rootstock vines in warm wet weather; sanitation of fallen debris	V3: thiophanate-methyl; V2 chlorothalonil (legacy)
Black Rot	Guignardia bidwellii (Fungal (warm, wet weather))	Destroy mummified berries; air circulation; copper on young bunches	None verified; V2 Bordeaux (legacy)
Bitter Rot	Greeneria uvicola (Fungal (field, storage and transit losses))	Prune canes and sanitise; no sourced weather thresholds	None verified (--)
Dead Arm (Phomopsis Cane and Leaf Spot)	Phomopsis viticola (Fungal (cane- and wood-borne))	Destroy pruned canes; prune back to healthy wood; Bordeaux paste on cuts	None verified; V2 Bordeaux paste

Quick separation of look-alikes: yellow oily spots on young leaves with white downy growth underneath, and shrivelled brown clusters, point to downy mildew; grey-white dusty coating on both leaf surfaces, corky or cracked berries, points to powdery mildew; small dark sunken berry lesions with dark margins (bird's-eye), shot-holed leaves and grey-black cane cankers point to anthracnose; water-soaked angular spots along veins on the underside of leaves in warm humid weather point to bacterial leaf spot; yellowish marginal leaf spots with concentric brown rings, plus dark patches on rachis and bunch stalk, point to Alternaria; grey fuzzy rot or watery rot appearing in storage points to Botrytis, Aspergillus or Rhizopus; orange powdery pustules under the leaf point to rust; reddish-brown leaf spots with black scabby, mummified berries point to black rot; water-soaked leaf lesions across veins with white-then-black canes point to bitter rot; and angular spots on canes that join into brown areas, with drying canes, point to dead arm.

Added entry (chapter 11): continues the summary table above

Disease	Pathogen / type	Main non-chemical measures	Chemical status
11. Esca (Black Measles)	Phaeomoniella chlamydospora, Phaeoacremonium spp. (Fungal; wood-infecting trunk-disease complex)	Protect pruning wounds; avoid wounding in wet weather; remove cankered wood in dry weather; healthy planting material	None verified (--); US wound protectants are not registered for esca in India

Esca (added): interveinal striping that dries and drops leaves, with small dark berry spots ringed brown-purple on the same shoots, and concentric dark spots in cut wood, point to esca.

1. Downy Mildew

Grape (Vitis vinifera) | Plasmopara viticola | Oomycete (fungus-like), rain-splash spread

At a glance. The most destructive grape disease in India. It attacks leaves, shoots, flowers, clusters and young berries, and spreads fast in wet, humid weather. Losses are highest when clusters are infected before fruit set. Sanitation at pruning, protectant sprays timed to growth stage and rotation of fungicide groups are the core of management.

1. Identification

Symptoms

- Crop / stage: leaves, shoot tips, flowers, clusters and young berries; the first symptoms appear at about the three-leaf stage after pruning (TB8).
- Leaves: light yellow spots on the upper surface of young mature leaves, with white spots on the lower surface; affected areas turn brown, and the leaf can no longer support bunch development (NHB). The early spots are greasy-looking 'oil spots' with white downy sporulation underneath in humid weather; severely infected leaves drop early (MDM, UDM).
- Shoots: shoot tips can become curled and thickened like a shepherd's crook (UDM).
- Clusters and berries: entire clusters can decay, dry and drop; infected small berries turn brown and mummify (NHB). Berries become resistant to infection about 3 weeks after bloom, but the rachis stays susceptible longer (MDM); once berries start softening and changing colour they no longer get infected (NHB).
- Losses are very high when clusters are attacked before fruit set (NHB).
- Further description (DP22): irregular, yellowish, translucent spots on the upper leaf surface with dirty white powdery fungal growth below; dwarfing of tender shoots; whitish growth on berries that later turn leathery and shrivel; infected berries turn hard, bluish-green and then brown (DP22).

Distinguishing features

- Often confused with powdery mildew. Downy mildew growth is mostly on the underside of the leaf, whereas powdery mildew growth is common on both leaf surfaces (UDM, VK).
- Downy mildew shows yellow oily spots on the upper surface with matching white growth below (NHB, MDM); powdery mildew forms a dusty grey-white coating on both surfaces (NHB).

2. Causes and progression

- Pathogen: Plasmopara viticola (NHB, TB8).
- Survival: thick-walled oospores in fallen leaves and fruit on the vineyard floor, which can stay viable for several years (VT, MDM). Downy-mildew-infected bunches left in the vineyard hold oospores in the dead tissue and are a primary source of inoculum (TB8). Dead wood on the vine can also harbour the pathogen (TB8).
- Primary infection: soil inoculum becomes active after a few major rains in June-July (TB8). Spores need a film of water to infect, and zoospores enter through stomata (MDM, VT).
- Secondary spread: sporangia form after humid nights and are spread by wind and rain splash; under ideal conditions a new generation can follow in 4-5 days (VT). Lesions show 5-17 days after infection (MDM).
- Indian crop cycle: after April foundation pruning the vine is in the monsoon vegetative phase; after October forward pruning the disease appears from the three-leaf stage and stays destructive until fruit set, about 50 days after pruning (TB8).
- Primary infection begins when spores are splashed from the soil to leaves; the fungus overwinters mainly in fallen leaves; secondary infection is by motile zoospores moved by splashing rain (DP22).
- Normally fully grown or maturing berries do not take fresh infection because stomata become non-functional; later berry infection gives soft-rot symptoms (DP22).

3. Favourable conditions

- Light continuous rain or heavy dew with high humidity and low temperatures (NHB).
- Free moisture on leaves or bunches for 2-3 hours after sunrise is needed for new infection (TB8).

- Night and morning temperatures of about 18-25 C (65-77 F) are ideal; limiting temperatures are below about 11 C (52 F) and above about 30 C (86 F) (VT; converted from degrees Fahrenheit).
- A common scouting rule for primary infection is 10 cm shoot growth, 10 mm rain and 10 C temperature in 24 hours (MDM; a Michigan guide).
- Gibberellic acid sprays increase shoot succulence and canopy humidity, which raises infection chances (TB8).
- After October pruning, rain, heavy dew and persistent fog at temperatures above 11 C favour disease development (DP22).

4. Prevention and management

- At and after harvest, remove abnormal unharvested bunches and burn them or dispose of them in a compost pit (TB8).
- Remove dead wood from arms and paste cut surfaces with Bordeaux paste, or spray 1% Bordeaux mixture within 1-2 days of pruning, because sap oozing from cuts gives spores a medium to germinate and infect within 2-3 days (TB8).
- Remove shoots emerging from the crown near the ground and tie up shoots hanging from the trellis during June, after monsoon onset (TB8).
- Collect infected leaves selectively before forward pruning and dispose of them in a compost pit (TB8). Do not bury downy-mildew leaves in the root zone in vineyards with heavy incidence (TB8).
- Prune after the second week of October to reduce damage, and remove and destroy all affected parts at pruning (NHB).
- Spray 0.5% Bordeaux mixture on an unpruned block next to a pruned block that will not be pruned within 5-8 days, well before new shoots appear in the pruned block (TB8).
- Do not use the same fungicide in successive sprays (TB8); avoid more than 2-3 sprays of systemic fungicides, because continuous systemic use encourages resistance and the resurgence of Alternaria and Botryodiplodia (NHB).
- When bunches are dipped in GA solution after fruit set, add a downy mildew fungicide at the spray concentration, so an infected bunch does not contaminate the dip and spread spores (TB8).
- NRCG has developed disease-forecasting systems that estimate disease risk at 3-day intervals; they have been installed at KVKs and co-operative societies, and spraying only when risk is predicted can cut the number of sprays (TB8).
- Further sanitation and canopy measures from international guides: prune for an open canopy and remove leaves around bunches soon after bloom (UDM).
- Keep the gap between pruning two adjacent blocks to no more than 9 days; if the gap is unavoidable, spray the unpruned block, because the pruned block receives inoculum from adjacent unpruned vines (DP22).
- Soil application of Trichoderma (2 L or kg per acre) or Bacillus (1 L or kg per acre), which can go through drip irrigation, supports induced resistance and reduces soil inoculum; bioagents help when fungicides are washed off in long rain (DP22). Bacillus subtilis 2 mL or g/L can be sprayed (DP22).
- Mineral spray oil at flowering helps reduce water retention in clusters during heavy rain and dew; monopotassium phosphate or SOP at 2-3 g/L helps reduce vigour and the development of downy and powdery mildew (DP22).

Regional note

Grape is grown mainly in Maharashtra (Nashik, Sangli, Solapur, Pune), Karnataka, Andhra Pradesh and Tamil Nadu. NRCG (Pune) guidance and its Indian label-claim list are the main sources used here.

5. Remedies and treatment

- Timing guidance from the ICAR-NRCG bulletin (TB8, dated 2007): after forward pruning, give the first protectant spray at the three-leaf stage (about 15 days after pruning) and two more at 5-10 day intervals around the 5 and 7 leaf stages, using different products; spray only if free moisture is present. Use a non-systemic fungicide 3-4 days after the second or third systemic spray if the gap between systemic sprays exceeds 10 days.

- In the monsoon season after foundation pruning, TB8 suggests Bordeaux mixture 0.5%, copper hydroxide or copper oxychloride at 15-20 day intervals, about three to four sprays from June to September. Its doses are older than the A5 list; where they differ, use the A5 dose below.
- Stage-wise guidance from DP22 (April 2022, reviewed by ICAR-NRC Grapes): at initiation of sprouting, spray or dust mancozeb 75 WP preventively if conditions are wet; at the three-leaf stage (11-15 days after pruning) and through 15-35 days, one of iprovalicarb + propineb, mandipropamid, ametoctradin + dimethomorph, cymoxanil + mancozeb, amisulbrom or cyazofamid; the total of these sprays should not exceed three in a season, and they should be applied before and after a rainy spell when humidity is relatively low, not during high humidity in the spell (DP22).
- If wet conditions last more than three days, prefer non-systemic dithiocarbamates (mancozeb 2 g/L or propineb 3 g/L) to prevent sporulation and secondary infection, and repeat a systemic fungicide once the rain stops (DP22). DP22 also prefers fosetyl-Al as a tank mix with mancozeb or propineb in wet weather; it quotes 4 g/L for this, which is higher than the A5 label-claim dose of 1.4-2.0 g/L, so follow the label.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Amisulbrom 17.7% SC w/w (20% SC w/v)	Downy mildew (NS)	375 mL/ha	1000 L/ha basis	V1	A5 PHI 59 d; label waiting period 59 d (DP22). *
Azoxystrobin 23% SC	Downy mildew (S)	494 mL/ha	1000 L/ha basis	V1	A5 PHI 7 d; label waiting period 7 d (DP22). *
Captan 50% WP	Downy mildew (NS)	2500 g/ha	1000 L/ha basis	V1	A5 PHI 70 d; label waiting period none stated (DP22)
Copper oxychloride 50% WP	Downy mildew (NS)	2.5 g/L (2.4 g/L as printed)	1000 L/ha basis	V1	A5 PHI 42 d; avoid after fruit set; label waiting period none stated (WP); 30 d for 50% WG (DP22)
Copper hydroxide 53.8% DF	Downy mildew (NS)	1.5 g/L	1000 L/ha basis	V1	A5 PHI 12 d; label waiting period 12 d (DP22)
Cyazofamid 34.5% SC	Downy mildew (NS)	200 mL/ha	1000 L/ha basis	V1	A5 PHI 30 d; label waiting period 7 d (DP22)
Dimethomorph 50% WP	Downy mildew (S)	0.50-0.75 g/L	1000 L/ha basis	V1	A5 PHI 34 d; label waiting period 25 d (DP22). *
Fosetyl-Al 80% WP	Downy mildew (S)	1.4-2.0 g/L	1000 L/ha basis	V1	A5 PHI 30 d; label waiting period 30 d (DP22)
Kresoxim-methyl 44.3% SC	Downy mildew (S)	600-700 mL/ha	1000 L/ha basis	V1	A5 PHI 30 d; label waiting period 7 d (DP22). *
Mancozeb 75% WP	Downy mildew (NS)	1.5-2.0 g/L	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period none stated (DP22)
Mandipropamid 23.4% SC	Downy mildew (NS)	0.8 mL/L	1000 L/ha basis	V1	A5 PHI 5 d; label waiting period 5 d (DP22). *
Metiram 70% WG	Downy mildew (NS)	2000 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 7 d (DP22). *
Propineb 70% WP	Downy mildew (NS)	3.0 g/L	1000 L/ha basis	V1	A5 PHI 75 d; avoid after fruit set; label waiting period 40 d (DP22)
Ametoctradin 27% + Dimethomorph 20.27% SC	Downy mildew (NS + S)	800-1000 mL/ha	1000 L/ha basis	V1	A5 PHI 34 d; label waiting period 34 d (DP22)
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Downy mildew (S + NS)	1500 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 21 d (DP22). *
Azoxystrobin 11% + Tebuconazole 18.3% w/w	Downy mildew (S + S)	750 mL/ha	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period 10 d (DP22). *
Benalaxyl-M 4% + Mancozeb 65% WP	Downy mildew (S + NS)	2750 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 48 d (DP22)

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Copper sulphate 47.15% + Mancozeb 30% WDG	Downy mildew (NS)	5000 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 10 d (DP22)
Copper sulphate pentahydrate 23.99% SC	Downy mildew (NS)	2.5 mL/L	1000 L/ha basis	V1	A5 PHI 30 d; label waiting period not in DP22 table (DP22)
Cymoxanil 8% + Mancozeb 64% WP	Downy mildew (S + NS)	2.0 g/L	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 10 d (DP22)
Dimethomorph 12% + Pyraclostrobin 6.7% WG	Downy mildew (S + S)	1500 g/ha	1000 L/ha basis	V1	A5 PHI 55 d; label waiting period 34 d (DP22). *
Famoxadone 16.6% + Cymoxanil 22.1% SC	Downy mildew (S + NS)	500 mL/ha	1000 L/ha basis	V1	A5 PHI 80 d; label waiting period 27 d (DP22). *
Fenamidone 10% + Mancozeb 50% WG	Downy mildew (S + NS)	2.5-3 g/L	1000 L/ha basis	V1	A5 PHI 85 d; label waiting period 85 d (DP22). *
Fenamidone 4.44% + Fosetyl-Al 66.66% WDG	Downy mildew (S)	2000-2500 g/ha	1000 L/ha basis	V1	A5 PHI 90 d; label waiting period 90 d (DP22). *
Fluopicolide 4.44% + Fosetyl-Al 66.67% WG	Downy mildew (S)	2.25-2.5 kg/ha	1000 L/ha basis	V1	A5 PHI 40 d; label waiting period 32 d (DP22)
Iprovalicarb 5.5% + Propineb 61.25% WP	Downy mildew (S + NS)	2.25 g/L	1000 L/ha basis	V1	A5 PHI 75 d; label waiting period not in DP22 table (DP22). *
Kresoxim-methyl 18% + Mancozeb 54% WP (72% WP)	Downy mildew (S + NS)	1500 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 5 d (DP22). *
Metalaxyl 8% + Mancozeb 64% WP	Downy mildew (S + NS)	2.5 g/L	1000 L/ha basis	V1	A5 PHI 75 d; label waiting period not less than 7 weeks (DP22)
Metalaxyl-M 4% + Mancozeb 64% WP	Downy mildew (S + NS)	2.5 g/L	1000 L/ha basis	V1	A5 PHI 75 d; label waiting period 8 d (DP22)
Metiram 44% + Dimethomorph 9% WG	Downy mildew (NS + S)	2500 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 10 d (DP22)
Oxathiapiprolin 3% + Mandipropamid 25% w/v (280 SC)	Downy mildew (S + NS)	700 mL/ha	1000 L/ha basis	V1	A5 PHI 30 d; label waiting period not in DP22 table (DP22)
Pyraclostrobin 5% + Metiram 55% 60 WG	Downy mildew (S + NS)	1.50-1.75 kg/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 34 d (DP22). *
Valifenalate 6% + Mancozeb 60% WG	Downy mildew (S + NS)	2000 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period not in DP22 table (DP22)
Aureofungin 46.15% w/v SP	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 15 d (DP22). Not on A5; check residue limits for export.
Captan 75% WP	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 8 d (DP22). Not on A5; check residue limits for export.

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Copper oxychloride 50% WG	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 30 d (DP22). Not on A5; check residue limits for export.
Copper sulphate 2.62% SC	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 3 d (DP22). Not on A5; check residue limits for export.
Chlorothalonil 75% WP	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 60 d (DP22). Not on A5; check residue limits for export.
Cymoxanil 50% WP	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 15 d (DP22). Not on A5; check residue limits for export.
Oxathiapiprolin 10.1% OD	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 5 d (DP22). Not on A5; check residue limits for export.
Picoxystrobin 22.52% w/w SC	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 7 d (DP22). Not on A5; check residue limits for export.
Zineb 75% WP	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period not stated (DP22). Not on A5; check residue limits for export.
Ziram 80% WP	Downy mildew	Per label (dose not given in DP22)	-	V3	Label waiting period none stated (DP22). Not on A5; check residue limits for export.

- * = fungicide group for which resistance has been detected in Indian grape-growing areas: QoI and CAA fungicides in downy mildew, and DMI (triazole) fungicides in powdery mildew (A5). A5 advises minimising formulations containing these fungicides and avoiding them in high-risk periods.
- Doses are for high-volume sprayers at a normal spray volume of 1000 L/ha. Volume can change with sprayer and canopy, but the amount of active ingredient per hectare should stay as recommended (A5). Systemic = S, non-systemic = NS (A5).
- Two waiting-period values are shown. 'A5 PHI' is the ICAR-NRCG export value, set with EU residue limits in mind and valid only if two applications are given per fruiting season, 7-15 days apart at the recommended dose (A5). 'Label waiting period' is the CIB&RC value reproduced in DP22 (2021 data) and is often shorter. If you export, use the stricter value. Neither replaces the current label.
- If a product proves ineffective, do not repeat it: repeated use raises residue risk and selects for resistance (A5).
- Copper fungicides: keep Bordeaux mixture at pH 7-7.5 and do not spray acidic materials after copper, because copper toxicity can cause chlorosis and necrosis (TB8).

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[TB8] ICAR-National Research Centre for Grapes - Technical Bulletin No. 8: Package of practices for managing major diseases and insect pests on grapes (March 2007). https://nrcgrapes.icar.gov.in/sites/default/files/publications/techbul-8-pop-for-disease-insect-pest-on-grapes.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[VK] Vikaspedia (IPM strategies for grapes) - Grapes: diseases and symptoms. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-diseases-and-symptoms

[MDM] Michigan State University IPM - Downy mildew (grape). https://www.canr.msu.edu/ipm/diseases/downy_mildew

[UDM] UC IPM - Downy mildew of grape (home and landscape). https://ipm.ucanr.edu/home-and-landscape/downy-mildew-of-grape/

[VT] Vineyard Team (UC IPM-based) - Downy mildew (Plasmopara viticola): damage, symptoms and life cycle. https://www.vineyardteam.org/resources/resource-library/pest-management.php?id=611

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

2. Powdery Mildew

Grape (Vitis vinifera) | Uncinula necator | Fungal (obligate parasite)

At a glance. Second most destructive grape disease after downy mildew and a key export concern, because it blemishes and deforms berries. It thrives in dense canopies in warm, dry or cloudy weather. Management combines canopy care, correctly timed systemic and non-systemic sprays, and strict rotation.

1. Identification

Symptoms

- Crop / stage: leaves, young shoots, blossoms and berries.
- White powdery (ash-like) coating in patches on both sides of leaves, young shoots and immature berries (NHB). Affected leaves turn pale and curl up; shoots stay weak and immature (NHB).
- Buds affected during the growing season may fail to sprout after October pruning, which reduces the number of productive canes (NHB).
- Blossoms that are infected fail to set fruit; young berries become corky; berries attacked at about 50% maturity turn dark and distorted, and severely affected berries are covered in white powder and crack (NHB).
- Yield loss comes from both berry drop and reduced berry size (NHB).
- Further description (DP22): infected spurs, canes, tendrils, panicles and berries have a white powdery appearance; patches of conidia and conidiophores may enlarge, coalesce and cover the whole leaf; the disease can attack at any stage of growth; floral infection causes flower shedding and poor fruit set, and early berry infection causes berry shedding (DP22).

Distinguishing features

- Easily confused with downy mildew; powdery mildew growth is common on both leaf surfaces, while downy mildew sporulation is mostly on the underside (UDM, VK).
- Powdery mildew on the rachis near maturity develops only when early infection during flowering to fruit set has occurred (TB8).

2. Causes and progression

- Pathogen: Uncinula necator (NHB, TB8, VK).
- Survival: the fungus overwinters in dormant buds or as specialised structures on the surface of the vines (VK).
- Progression: it can infect all green tissues of the vine (VK). Buds infected during the season carry the disease into the next cycle and may fail to sprout (NHB).
- Survival (DP22): the pathogen survives off-season as mycelium in dormant buds; when conditions are favourable, shoots from infected buds become covered with white mycelium, and wind blows conidia from them to cause secondary infection (DP22).

3. Favourable conditions

- Warm, dry conditions; shade or diffused light also help (NHB).
- Dense canopy in summer, especially after flood irrigation (TB8).
- Cloudy weather with little or no rain (under 2 mm) for 3-5 days during June-August (TB8).
- Around 50 days after forward pruning, falling temperatures, dense canopy and cloudy, low-light days with relatively warm temperatures (20-25 C) are most favourable (TB8).
- Potassium deficiency curls leaves upward, creating a microclimate that favours the fungus (TB8).
- A warm winter in December-January, with temperatures of 20-33.5 C, is very favourable; low, diffused light favours the disease, so severity can be greater inside the canopy (DP22).

4. Prevention and management

- Keep the canopy open and avoid excessive density, especially in summer and after flood irrigation (TB8).

- Maintain adequate potassium nutrition. TB8 suggests potassium sources (potassium nitrate, potassium sulphate or monopotassium phosphate) at 2-3 g/L between systemic sprays at 70-80 days after pruning, with no more than one monopotassium phosphate spray because it can reduce bloom on bunches (TB8).
- Do not spray sulphur when temperatures are very high (35-40 C) because of phytotoxicity, and avoid sulphur after fruit set because it can stain berries (TB8). Take care when spraying sulphur, since it can scorch berry skin and leave black specks (NHB).
- Do not repeat the same fungicide during the fruit-set to veraison window; the PHI scheme in TB8 spaces four systemic sprays 10 days apart from about 50 to 90 days after pruning, with the PHI and expected harvest date guiding the sequence (TB8).
- In between two systemic sprays use a non-systemic fungicide or sulphur to avoid resistance (NHB, TB8). No single chemical should be sprayed more than twice (NHB).
- Resistance to triazole (DMI) fungicides in powdery mildew has been detected in India, so these products should be minimised (A5).
- Spray coverage of the inner canopy is essential, and the 15-35 day stage after pruning is when it can be ensured (DP22).
- Soil application of Bacillus (1 L or kg per acre) a week before pruning and again at the dormant-to-swollen bud stage, for induced resistance (DP22). Bacillus subtilis 2 mL or g/L can be sprayed from 15-35 days, and at 50-60 days if minimum temperature is above 10 C and relative humidity above 60%; alternate it with fungicides, leaving 48-72 hours between sprays (DP22).
- After veraison, DP22 lists sulphur 80 WG 2.0 g/L or Bacillus subtilis, and, if active mildew growth is noticed, low-residue options such as silver complex of hydrogen peroxide or chlorine dioxide, noting that these also kill biocontrol agents (DP22). Trichoderma 5 mL or g/L is sprayed 20 days before harvest, and Bacillus subtilis or Trichoderma 10 days before harvest (DP22).

Regional note

Disease timing in the NRCG bulletin follows Maharashtra's two cycles (foundation pruning in April, forward pruning in October); in other states, follow the local SAU / KVK schedule.

5. Remedies and treatment

- TB8 (2007) also suggests, for powdery mildew noticed in the last week of May or in cloudy weather in June-August, sulphur 2.0 g/L or potassium bicarbonate 5-10 g/L (V2 rows below). Do not spray potassium bicarbonate after veraison because bunches can turn yellowish and mature early (TB8).
- Stage-wise guidance from DP22: at 15-35 days after pruning, one of hexaconazole 5 EC (1 mL/L), difenoconazole 25 EC (0.5 mL/L), fluopyram + tebuconazole (0.5 mL/L), myclobutanil 10 WP (0.4 g/L) or tetraconazole 3.8 EW (0.75 mL/L); at 36-50 days (flowering to fruit set) the same list plus pyraclostrobin + fluxapyroxad (0.2 mL/L), metrafenone 50 SC (0.25 mL/L) or cyflufenamid 5% EW (0.5 mL/L); at 60-90 days sulphur 80 WG (2.0 g/L) or the products above, chosen by their PHI in A5 (DP22). These doses agree with the A5 label-claim list.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 23% SC	Powdery mildew (S)	494 mL/ha	1000 L/ha basis	V1	A5 PHI 7 d; label waiting period 7 d (DP22). *
Bupirimate 25% w/v (26.7% w/w) SC	Powdery mildew (S)	3.0 mL/L	1000 L/ha basis	V1	A5 PHI 45 d; label waiting period not in DP22 table (DP22)
Cyflufenamid 5% EW	Powdery mildew (S)	500 mL/ha	1000 L/ha basis	V1	A5 PHI 50 d; label waiting period 25 d (DP22)
Difenoconazole 25% EC	Powdery mildew (S)	0.50 mL/L	1000 L/ha basis	V1	A5 PHI 45 d; label waiting period 10 d (DP22). *
Flusilazole 40% EC	Powdery mildew (S)	25 mL/200 L	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period not in DP22 table (DP22). *
Hexaconazole 5% EC	Powdery mildew (S)	1.0 mL/L	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period 14 d (DP22). *
Kresoxim-methyl 44.3% SC	Powdery mildew (S)	600-700 mL/ha	1000 L/ha basis	V1	A5 PHI 30 d; label waiting period 7 d (DP22). *

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Meptyldinocap 35.7% EC	Powdery mildew (NS)	308.6-342.8 mL/ha	1000 L/ha basis	V1	A5 PHI 50 d; label waiting period 30 d (DP22)
Metrafenone 50% SC	Powdery mildew (S)	250 mL/ha	1000 L/ha basis	V1	A5 PHI 22 d; label waiting period 22 d (DP22)
Myclobutanil 10% WP	Powdery mildew (S)	0.40 g/L	1000 L/ha basis	V1	A5 PHI 30 d; label waiting period 15 d (DP22). *
Penconazole 10% EC	Powdery mildew (S)	0.50 mL/L	1000 L/ha basis	V1	A5 PHI 50 d; label waiting period 30 d (DP22). *
Polyoxin D zinc salt 5% SC	Powdery mildew (S)	600 mL/ha	1000 L/ha basis	V1	A5 PHI 15 d; label waiting period not in DP22 table (DP22)
Sulphur 40% SC / 55.16% SC / 80% WP / 80% WDG / 85% WP	Powdery mildew (NS)	3.0 mL, 3.0 mL, 2.50 g, 1.87-2.50 g, 1.50-2.0 g per L respectively	1000 L/ha basis	V1	A5 PHI not applicable; label waiting period 10 d for 55.16% SC; none stated for WP, DP and WDG (DP22)
Sulphur 58.5% w/w + Azoxystrobin 4.2% w/w SC	Powdery mildew (S)	2.0 mL/L	1000 L/ha basis	V1	A5 PHI 7 d; label waiting period not in DP22 table (DP22)
Tetraconazole 3.8% EW	Powdery mildew (S)	0.75 mL/L	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period 30 d (DP22). *
Azoxystrobin 11% + Tebuconazole 18.3% w/w	Powdery mildew (S + S)	750 mL/ha	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period 10 d (DP22). *
Azoxystrobin 25% + Boscalid 35% WG	Powdery mildew (S + S)	500 g/ha	1000 L/ha basis	V1	A5 PHI 7 d; label waiting period not in DP22 table (DP22). *
Boscalid 25.2% + Pyraclostrobin 12.8% w/w WG	Powdery mildew (S + S)	500-600 g/ha	1000 L/ha basis	V1	A5 PHI 55 d; label waiting period 34 d (DP22). *
Fluopyram 200 + Tebuconazole 200 SC	Powdery mildew (S + S)	0.563 mL/L	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period 10 d (DP22). *
Fluxapyroxad 25% + Pyraclostrobin 25% SC	Powdery mildew (S + S)	200 mL/ha	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period 10 d (DP22)
Fluxapyroxad 75 g/L + Difenoconazole 50 g/L SC	Powdery mildew (S + S)	800 mL/ha	1000 L/ha basis	V1	A5 PHI 45 d; label waiting period 32 d (DP22). *
Tebuconazole 50% + Trifloxystrobin 25% WG	Powdery mildew (S + S)	0.175 g/L	1000 L/ha basis	V1	A5 PHI 34 d; label waiting period 34 d (DP22). *
Potassium bicarbonate	Powdery mildew (S)	5.0 g/L alone or with a systemic fungicide (10.0 g/L in cloudy weather), with a non-ionic spreader	1000 L/ha basis	V2	TB8 only. Not after veraison. Registration for grape not matched in A5.
HP grape spray oil (mineral oil) or azadirachtin 0.03%	Powdery mildew on rachis (NS)	Oil 5-7 mL/L or azadirachtin 5 mL/L	400 L/ha (full canopy) per TB8	V2	TB8 only. For rachis infection when fewer than 30 days remain to harvest; test on a few bunches first for spots or scorching.
Carbendazim 46.27% SC	Powdery mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 30 d (DP22). Not on A5; check residue limits for export.
Lime sulphur 22% SC	Powdery mildew	Per label (dose not given in DP22)	-	V3	Label waiting period none stated (DP22). Not on A5; check residue limits for export.
Picoxystrobin 22.52% w/w SC	Powdery mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 7 d (DP22). Not on A5; check residue limits for export.

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiophanate-methyl 70% WP	Powdery mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 7 d (DP22). Not on A5; check residue limits for export.
Triadimefon 25% WP	Powdery mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 25 d (DP22). Not on A5; check residue limits for export.
Hexaconazole 2% SC and 5% SC	Powdery mildew	Per label (dose not given in DP22)	-	V3	Label waiting period 14 d (DP22). Not on A5; check residue limits for export.

- * = fungicide group for which resistance has been detected in Indian grape-growing areas: QoI and CAA fungicides in downy mildew, and DMI (triazole) fungicides in powdery mildew (A5). A5 advises minimising formulations containing these fungicides and avoiding them in high-risk periods.
- Doses are for high-volume sprayers at a normal spray volume of 1000 L/ha. Volume can change with sprayer and canopy, but the amount of active ingredient per hectare should stay as recommended (A5). Systemic = S, non-systemic = NS (A5).
- Two waiting-period values are shown. 'A5 PHI' is the ICAR-NRCG export value, set with EU residue limits in mind and valid only if two applications are given per fruiting season, 7-15 days apart at the recommended dose (A5). 'Label waiting period' is the CIB&RC value reproduced in DP22 (2021 data) and is often shorter. If you export, use the stricter value. Neither replaces the current label.
- If a product proves ineffective, do not repeat it: repeated use raises residue risk and selects for resistance (A5).
- Dinocap in TB8 and the older NHB product names (Bayleton, Calaxin, benomyl) are not on the current A5 list and are not reproduced; meptyldinocap and triazoles on the A5 list apply instead.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[TB8] ICAR-National Research Centre for Grapes - Technical Bulletin No. 8: Package of practices for managing major diseases and insect pests on grapes (March 2007). https://nrcgrapes.icar.gov.in/sites/default/files/publications/techbul-8-pop-for-disease-insect-pest-on-grapes.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[VK] Vikaspedia (IPM strategies for grapes) - Grapes: diseases and symptoms. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-diseases-and-symptoms

[UDM] UC IPM - Downy mildew of grape (home and landscape). https://ipm.ucanr.edu/home-and-landscape/downy-mildew-of-grape/

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

3. Anthracnose (Bird's-eye Rot)

Grape (Vitis vinifera) | Elsinoe ampelina | Fungal (cane- and debris-borne; spread by rain and dew)

At a glance. A disease of tender shoots, young leaves, flowers and berries that is prevalent in all Indian grape regions. Infections need wet conditions and only affect tender tissue, so sanitation after pruning and protective sprays at the tender-growth stage matter most.

1. Identification

Symptoms

- Crop / stage: tender shoots, young leaves, flowers, tendrils, petioles, leaf veins, fruit stems and young berries (NHB, VK).
- Small light brown or greyish-black lesions on tender shoots, young leaves, flowers and young berries; short holes in leaves reduce effective leaf area; affected blossoms fail to set fruit (NHB).
- Cankers on petioles and veins, and twisted, deformed leaves (NHB).
- Berries: circular brown sunken spots with dark brown margins ('bird's-eye' appearance); severely affected berries may crack and expose the seeds; bunches lose export value (NHB, VK). Infected grapes often shatter, leaving only the stem (VK).
- Further description (DP22): on leaves, small round spots with a hole at the centre (shot-hole); in severe infection leaves shrivel and drop. On shoots, deep elongated cankers, greyish in the centre with a black edge. Inflorescences are highly susceptible and can turn yellow, brown and dry out completely (DP22).

Distinguishing features

- Berry lesions with a dark margin and sunken pale centre give the 'bird's-eye rot' look (VK, NHB).
- Anthracnose is one of three diseases that commercial varieties are highly susceptible to in India, along with downy and powdery mildew (KR).

2. Causes and progression

- Pathogen: Elsinoe ampelina (NHB, TB8).
- Survival and spread: spores can develop on infected pruned cane in wet conditions and re-infect new shoots, so pruned material left near the vineyard, or open on the soil surface, is a source of inoculum (TB8).
- Progression: new infection occurs only on tender shoots and leaves, in wet conditions (TB8).
- Survival (DP22): in cankers on canes and in infected berries on the ground or left on the trellis; in spring small fruiting bodies (acervuli) produce conidia that stick to the site of infection and are dispersed by rain (DP22).

3. Favourable conditions

- Rain and dew are highly favourable for spread (NHB).
- The disease is more problematic during October and November (NHB), and after summer showers that follow bud emergence (TB8).
- The longer leaves and stems stay wet (more than 12 hours), the more severe the infection; the disease develops at 10-35 C with an optimum of 20-26 C, and symptoms show 4-12 days after infection (DP22).
- It is extremely severe at the cane-maturity stage (61-90 days after foundation pruning), when the vines receive intermittent showers and drizzles (DP22).

4. Prevention and management

- Remove all pruned material from the vineyard soon after pruning and dispose of it in a compost pit; do not leave it near the vineyard during rains (TB8). All twigs or canes with cankers should be removed at pruning and the pruned material burnt or buried deep (NHB).
- Mix mancozeb (6.0 g/L) or carbendazim (2.0 g/L) with the hydrogen cyanamide paste used on canes, to kill inoculum left on the cane (TB8, V2 practice).
- Spray protectively on new shoots and young branches at the 3-4 leaf stage (NHB). Copper sprays suggested for downy mildew also control anthracnose (TB8).

- Carbendazim is prone to resistance, so follow it 7-10 days later with a non-systemic fungicide (TB8). Carbendazim breaks down in alkaline water: spray water should be below pH 7, and 25-30 g citric acid per 100 L can be added if the water is above pH 7.5; add the citric acid before the carbendazim, not after (TB8).
- Spray Trichoderma formulations 2-5 mL/L, one or two sprays 10 days apart in high-humidity September-October weather, to limit Alternaria and Cladosporium infection of leaves and delay leaf fall; avoid fungicide sprays after fungal biocontrol sprays (TB8).
- At initiation of sprouting, if it rains, protect young tips with carbendazim 50 WP (1.0 g/L) or thiophanate-methyl 70 WP (0.75 g/L); at the cane-maturity stage thiophanate-methyl or carbendazim at 1 g/L twice at weekly intervals gives good control (DP22).
- Dithiocarbamates and copper sprays used for downy mildew also control anthracnose and no separate spray may be needed; if incidence appears, difenoconazole 25 EC at 0.5 mL/L can be used, and it also reduces shoot vigour and downy mildew risk (DP22).
- Mancozeb 75 WP (5 g/L) mixed with hydrogen cyanamide (30-40 mL/L) for cane swabbing helps kill inoculum on canes (DP22).

Regional note

Anthracnose is described as prevalent in all grape-growing regions of the country (NHB).

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Carbendazim 50% WP / 46.27% SC	Anthracnose (S)	1.0 g/L or 1.0 mL/L	1000 L/ha basis	V1	A5 PHI 50 d; label waiting period none stated (50% WP) (DP22)
Copper oxychloride 50% WP	Anthracnose (NS)	2.5 g/L (2.40 g/L as printed)	1000 L/ha basis	V1	A5 PHI 42 d; avoid after fruit set; label waiting period none stated (DP22)
Propineb 70% WP	Anthracnose (NS)	3.0 g/L	1000 L/ha basis	V1	A5 PHI 75 d; label waiting period not in DP22 table (DP22)
Thiophanate-methyl 70% WP	Anthracnose (S)	0.71-0.95 g/L	1000 L/ha basis	V1	A5 PHI 73 d; label waiting period 7 d (DP22)
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Anthracnose (S + NS)	1500 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 21 d (DP22)
Carbendazim 12% + Mancozeb 63% WP	Anthracnose (S + NS)	1500 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 7 d (DP22)
Copper sulphate 47.15% + Mancozeb 30% WDG	Anthracnose (NS + NS)	5000 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 10 d (DP22)
Fluopyram 200 + Tebuconazole 200 SC	Anthracnose (S + S)	0.563 mL/L	1000 L/ha basis	V1	A5 PHI 60 d; label waiting period 10 d (DP22)
Kasugamycin 5% + Copper oxychloride 45% WP	Anthracnose (S + NS)	750 g/ha	1000 L/ha basis	V1	A5 PHI 70 d; avoid after flowering; label waiting period 37 d (DP22)
Kresoxim-methyl 18% + Mancozeb 54% WP (72% WP)	Anthracnose (S + NS)	1500 g/ha	1000 L/ha basis	V1	A5 PHI 66 d; label waiting period 5 d (DP22)
Sulphur 58.5% w/w + Azoxystrobin 4.2% w/w SC	Anthracnose (S)	2.0 mL/L	1000 L/ha basis	V1	A5 PHI 7 d; label waiting period not in DP22 table (DP22)
Aureofungin 46.15% w/v SP	Anthracnose	Per label (dose not given in DP22)	-	V3	Label waiting period 15 d (DP22). Not on A5; check residue limits for export.

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Difenoconazole 25% EC	Anthracnose	Per label (dose not given in DP22)	-	V3	Label waiting period 10 d (DP22). Not on A5; check residue limits for export.
Iprodione 50% WP	Anthracnose	Per label (dose not given in DP22)	-	V3	Label waiting period 20 d (DP22). Not on A5; check residue limits for export.
Kitazin (IBP) 48% EC	Anthracnose	Per label (dose not given in DP22)	-	V3	Label waiting period 15 d (DP22). Not on A5; check residue limits for export.
Mancozeb 75% WP	Anthracnose	Per label (dose not given in DP22)	-	V3	Label waiting period none stated (DP22). Not on A5; check residue limits for export.
Ziram 80% WP	Anthracnose	Per label (dose not given in DP22)	-	V3	Label waiting period none stated (DP22). Not on A5; check residue limits for export.
Chlorothalonil 75% WP	Anthracnose	Per label (dose not given in DP22)	-	V3	Label waiting period 60 d (DP22). Not on A5; check residue limits for export.

- Doses are for high-volume sprayers at a normal spray volume of 1000 L/ha. Volume can change with sprayer and canopy, but the amount of active ingredient per hectare should stay as recommended (A5). Systemic = S, non-systemic = NS (A5).
- Two waiting-period values are shown. 'A5 PHI' is the ICAR-NRCG export value, set with EU residue limits in mind and valid only if two applications are given per fruiting season, 7-15 days apart at the recommended dose (A5). 'Label waiting period' is the CIB&RC value reproduced in DP22 (2021 data) and is often shorter. If you export, use the stricter value. Neither replaces the current label.
- If a product proves ineffective, do not repeat it: repeated use raises residue risk and selects for resistance (A5).
- Doses for copper and carbendazim in TB8 (for example copper oxychloride 3.0 g/L) differ from the current A5 list; use the A5 values.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[TB8] ICAR-National Research Centre for Grapes - Technical Bulletin No. 8: Package of practices for managing major diseases and insect pests on grapes (March 2007). https://nrcgrapes.icar.gov.in/sites/default/files/publications/techbul-8-pop-for-disease-insect-pest-on-grapes.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[VK] Vikaspedia (IPM strategies for grapes) - Grapes: diseases and symptoms. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-diseases-and-symptoms

[KR] KRISHI (ICAR) - Sawant, I.S., Sawant, S.D. and Saha: Integrated disease management strategies in grapes (2016), abstract. https://krishi.icar.gov.in/jspui/handle/123456789/64829

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

4. Bacterial Leaf Spot

Grape (Vitis vinifera) | Xanthomonas campestris pv. viticola | Bacterial (spread in warm, humid weather)

At a glance. A bacterial disease of leaves, shoots and berries that is gaining importance as a minor but increasing problem in Indian vineyards. Warm, humid weather favours it; sanitation and protectant sprays are the tools, and only one combination product has a verified label claim.

1. Identification

Symptoms

- Crop / stage: young growing shoots are affected first; leaves, shoots and berries (NHB).
- Minute water-soaked spots on the lower surface of leaves along the main and lateral veins; the spots later coalesce into larger patches (NHB).
- Berries: brownish-black lesions; affected berries become small and shrivelled (NHB).
- In severe cases the spots enlarge and become angular, and leaves can be completely defoliated (DP22).

Distinguishing features

- The DPPQS IPM package reviewed by ICAR-NRC Grapes names the pathogen as Xanthomonas campestris pv. viticola (DP22), whereas the NIPHM pest list gives pv. campestris (NG, NP) and NHB gives only Xanthomonas campestris (NHB). The pv. viticola name is used here because it comes from the NRCG-reviewed package.
- Alternaria leaf blight also begins as yellowish spots on leaves; its spots enlarge into brown patches with concentric rings (NHB; see chapter 5).

2. Causes and progression

- Pathogen: Xanthomonas campestris (NHB, NP).
- Spread: collecting and burning infected plant parts is advised to limit spread (NHB), which indicates infected debris carries the bacterium.
- Importance: ICAR-NRCG scientists describe rust and bacterial leaf spot as minor diseases that are gaining importance, mainly due to climate change (KR).
- Survival (DP22): the pathogen survives in infected plant residue in soil and is transmitted through propagative material.

3. Favourable conditions

- Temperatures of 25-30 C with relative humidity of 80-90% (NHB).
- Prevalent during June-August and again in February-March (NHB).

4. Prevention and management

- Collect and burn infected plant parts to minimise spread (NHB); use healthy propagative material, since the bacterium is carried on it (DP22).
- Mancozeb applied for downy mildew also controls bacterial spot; copper oxychloride or copper hydroxide can control the disease effectively; where anthracnose and bacterial spot occur together, kasugamycin 5% + copper oxychloride 45% at 0.75 g/L gives good results (DP22).
- NHB suggests weekly copper fungicide or Bordeaux mixture sprays from the last week of October as prevention (NHB, older publication).

Regional note

No state-specific recommendation (for example from MPKV Rahuri or ICAR-NRCG beyond A5) was found for this disease.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Kasugamycin 5% + Copper oxychloride 45% WP	Bacterial leaf spot (S + NS)	750 g/ha (0.75 g/L at 1000 L/ha)	1000 L/ha basis	V1	A5 PHI 70 d; avoid after flowering. Label waiting period 37 d (DP22).
Copper oxychloride or copper hydroxide (preventive)	Bacterial leaf spot	Copper oxychloride 3.0 g/L or copper hydroxide 2.0 g/L as used for downy mildew (DP22)	-	V2	DP22 says these 'can control' the disease; no single-copper bacterial-leaf-spot label claim matched. A5 gives copper oxychloride 2.5 g/L for downy mildew, so follow the label.
Mancozeb 75 WP	Bacterial leaf spot (NS)	2 g/L (as used for downy mildew)	-	V2	DP22 notes it controls bacterial spot as a side effect of downy mildew sprays; no bacterial-leaf-spot label claim matched.

- Doses are for high-volume sprayers at a normal spray volume of 1000 L/ha. Volume can change with sprayer and canopy, but the amount of active ingredient per hectare should stay as recommended (A5). Systemic = S, non-systemic = NS (A5).
- Two waiting-period values are shown. 'A5 PHI' is the ICAR-NRCG export value, set with EU residue limits in mind and valid only if two applications are given per fruiting season, 7-15 days apart at the recommended dose (A5). 'Label waiting period' is the CIB&RC value reproduced in DP22 (2021 data) and is often shorter. If you export, use the stricter value. Neither replaces the current label.
- If a product proves ineffective, do not repeat it: repeated use raises residue risk and selects for resistance (A5).
- Streptocycline (500 ppm), which NHB lists as a prophylactic spray, is not on the A5 label-claim list and is not reproduced.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[NP] NIPHM and Directorate of Plant Protection, Quarantine & Storage - Grapes pests list (via Vikaspedia). https://en.vikaspedia.in/viewcontent/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-pests?lgn=en

[KR] KRISHI (ICAR) - Sawant, I.S., Sawant, S.D. and Saha: Integrated disease management strategies in grapes (2016), abstract. https://krishi.icar.gov.in/jspui/handle/123456789/64829

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

[NG] NIPHM and DPPQS - AESA based IPM package for Grapes (pest list: diseases of national and regional significance). https://niphm.gov.in/IPMPackages/Grapes.pdf

5. Alternaria Leaf Blight and Bunch Necrosis

Grape (Vitis vinifera) | Alternaria alternata (also A. tenuissima) | Fungal (weak pathogen; often follows heavy systemic fungicide use)

At a glance. A leaf and bunch disease that appears in June and December, drying and defoliating leaves and causing dark patches on berries, rachis and bunch stalk. It can continue as rot in transit and storage. No chemical is on the current label-claim list, so management relies on sanitation, avoiding resistance-driven resurgence and biological control.

1. Identification

Symptoms

- Crop / stage: leaves, berries, rachis and bunch stalk (NHB).
- Small yellowish spots first appear along the leaf margins and enlarge into brownish patches with concentric rings; severe infection dries and defoliates the leaves (NHB).
- Dark brown-purplish patches on infected berries, on the rachis and on the bunch stalk just below its attachment to the shoot (NHB).
- If berry infection is not controlled in the field it can lead to berry rotting during transit and storage (NHB).

Distinguishing features

- The early yellowish marginal leaf spots resemble early bacterial leaf spot, but Alternaria patches develop concentric rings (NHB).
- NIPHM lists both Alternaria alternata and A. tenuissima as causes of Alternaria blight in grape (NP).

2. Causes and progression

- Pathogens: Alternaria alternata and A. tenuissima (NP, NHB).
- Progression: the disease occurs in June and again in December, spreading from leaves to fruit (NHB).
- Resurgence: continuous use of systemic fungicides for downy mildew encourages resistant pathogen strains and the resurgence of secondary diseases such as Alternaria and Botryodiplodia (NHB).
- Survival (DP22): through conidia or mycelium in diseased plant debris or weeds; it appears in June and December.

3. Favourable conditions

- The disease is recorded in June and December (NHB), the monsoon vegetative phase and the winter crop phase. Specific temperature and humidity thresholds were not found in the sources reviewed.
- Trichoderma sprays are used in humid September-October weather to limit Alternaria and Cladosporium infection of leaves (TB8), indicating that high humidity at that stage favours these fungi.
- Overuse of systemic fungicides is a driver (NHB).

4. Prevention and management

- Limit systemic sprays to 2-3 per season, rotate chemical groups and use non-systemic partners (NHB, TB8).
- Spray Trichoderma formulations 2-5 mL/L, one or two sprays 10 days apart during high humidity in September-October, to limit Alternaria and Cladosporium and delay leaf fall (TB8).
- After pruning, remove pruned material and infected leaves; bury removed leaves in the root zone to raise soil Trichoderma, except in vineyards with heavy downy mildew incidence (TB8).
- Do not spray fungicides after fungal biocontrol sprays, because Trichoderma on foliage will be killed (TB8).

Regional note

No state-specific recommendation was found; the NHB and ICAR-NRCG material is for all-India or Maharashtra conditions.

5. Remedies and treatment

- No chemical on the ICAR-NRCG list of CIB&RC label-claim chemicals (A5) is listed for Alternaria. NHB (a legacy publication) names Bordeaux mixture, mancozeb, thiophanate-methyl, ziram and captan alternated weekly in June-August and from December to harvest (NHB). Mancozeb and captan have grape label claims for downy mildew in A5, but no Alternaria label claim was matched, so none is recommended here.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Trichoderma liquid formulation	Alternaria / Cladosporium on leaves (biological)	2-5 mL/L; one or two sprays 10 days apart	1000 L/ha basis (400 L/ha for full canopy per TB8)	V2	TB8 only. Spray in the evening; biocontrol registration status not verified in this review.

- No chemical is recommended for Alternaria because no label claim was matched in A5. Confirm any option with ICAR-NRCG, the state horticulture department or the nearest KVK.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[TB8] ICAR-National Research Centre for Grapes - Technical Bulletin No. 8: Package of practices for managing major diseases and insect pests on grapes (March 2007). https://nrcgrapes.icar.gov.in/sites/default/files/publications/techbul-8-pop-for-disease-insect-pest-on-grapes.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[NP] NIPHM and Directorate of Plant Protection, Quarantine & Storage - Grapes pests list (via Vikaspedia). https://en.vikaspedia.in/viewcontent/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-pests?lgn=en

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

6. Post-harvest Berry Rots (Botrytis, Aspergillus, Rhizopus and others)

Grape (Vitis vinifera) | Botrytis cinerea, Aspergillus niger, Rhizopus sp., Cladosporium, Alternaria, Botryodiplodia | Fungal (wound-entry rots in transit and storage)

At a glance. A group of fungal rots that start in the vineyard or at packing and show up in transit and storage. Careful handling, clean packing and prompt cooling matter most, backed by pre-harvest measures.

1. Identification

Symptoms

- Botrytis grey mould: the skin of affected berries just below the infection becomes loose and slips off when rubbed, leaving the pulp exposed; infected berries shrivel, rot and turn dark brown with a greyish fungal growth; in the vineyard it can attack shoots and clusters or destroy stalks, leading to fruit drop (NHB).
- Aspergillus niger black rot: pulp of infected berries is reduced to watery consistency (NHB).
- Rhizopus rot: under warm, moist conditions the fungus quickly forms a coarse grey mat of mycelium (NHB).
- The NIPHM list also names blue mould (Penicillium), black mould (A. niger) and green mould (A. flavus) as post-harvest berry rots (NP).

Distinguishing features

- Botrytis can grow at low temperature, so it is one of the most important storage diseases (NHB).
- Aspergillus and Rhizopus rots enter through injuries from poor handling or tight packing (NHB).

2. Causes and progression

- Pathogens: the TB8 bulletin names Cladosporium, Alternaria, Botryodiplodia, Rhizopus and Aspergillus as post-harvest berry rot fungi (TB8); NHB adds Botrytis cinerea (NHB).
- Entry: Aspergillus enters berries through injuries caused by poor post-harvest handling; Rhizopus is helped by injury from tight packing and by storage temperature (NHB).
- Progression: if infected berries are trimmed at harvest, Rhizopus rot does not occur after harvest under ideal storage conditions (NHB).

3. Favourable conditions

- High storage temperature and humid conditions favour Aspergillus black rot (NHB).
- Warm and moist conditions favour Rhizopus (NHB).
- Humidity around clusters in the vineyard favours Botrytis (NHB).

4. Prevention and management

- Careful handling in the field, precooling and refrigeration (NHB).
- Prune and thin the vineyard to reduce humidity around clusters (NHB).
- Prompt refrigeration at 1-2 C or below prevents Aspergillus rot (NHB); cold storage at 0-1 C prevents Rhizopus growth (NHB).
- SO2-releasing pads in the packing boxes help control black rot and Rhizopus; remove diseased berries during grading and avoid injury while packing (NHB).
- Two Trichoderma sprays about 15 days before harvest at weekly interval help control post-harvest berry rots and also help with bio-remediation of pesticide residue on berries (TB8).
- Harvest early in the morning and stop before the temperature reaches 20 C; use clean, light-coloured crates holding two layers of bunches, keep filled crates in shade, and grade and pack at about 15-17 C (DP22).
- Take grapes to pre-cooling within 4-6 hours of harvest and bring the pulp to 1-3 C; then place dual-release SO2 pads on the filled pouches and store at 0 +/- 0.5 C and 93 +/- 2% relative humidity; table grapes keep 2-3 months under this cold storage, and any break in the cold chain leads to quality loss and post-harvest disease (DP22).
- Spray Trichoderma (5 mL or g/L) 20 days before harvest and Bacillus subtilis (2 mL or g/L) or Trichoderma 10 days before harvest, which also help degrade pesticide residues on berries (DP22).

Regional note

No state-specific recommendation was found; handling and cold-chain measures apply to all regions.

5. Remedies and treatment

- The current ICAR-NRCG label-claim list (A5) has no entry for post-harvest berry rots. The 2007 NRCG bulletin (TB8) suggests the two V2 options below; neither is on A5.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Iprodione	Post-harvest berry rots	2 g/L, 7-15 days before harvest	400 L/ha (full canopy) per TB8	V2	TB8 only (2007). PHI 7 d per TB8; not on A5. Do not spray within 5 days before or after Trichoderma. Confirm the current label before use.
Trichoderma liquid formulation	Post-harvest berry rots (biological)	5 mL/L; two sprays about 15 days before harvest, weekly interval	400 L/ha (full canopy) per TB8	V2	TB8 only. Preponed in rainy weather. Biocontrol registration status not verified in this review. DP22 (2022) also gives Trichoderma 5 mL or g/L at 20 and 10 days before harvest.

- NHB (legacy) also lists captan and benomyl/carbendazim as pre-harvest prophylactic sprays for Botrytis and Rhizopus; these are not reproduced because no post-harvest rot label claim was matched on A5.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[TB8] ICAR-National Research Centre for Grapes - Technical Bulletin No. 8: Package of practices for managing major diseases and insect pests on grapes (March 2007). https://nrcgrapes.icar.gov.in/sites/default/files/publications/techbul-8-pop-for-disease-insect-pest-on-grapes.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[NP] NIPHM and Directorate of Plant Protection, Quarantine & Storage - Grapes pests list (via Vikaspedia). https://en.vikaspedia.in/viewcontent/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-pests?lgn=en

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

7. Leaf Rust

Grape (Vitis vinifera) | Phakopsora vitis | Fungal (rust; wind-borne urediniospores)

At a glance. A regional leaf disease marked by orange pustules under the leaf and premature leaf fall. It is generally not reported on commercial table and wine varieties in Maharashtra but is seen on rootstocks, and has been important on Bangalore Blue in Karnataka.

1. Identification

Symptoms

- Crop / stage: leaves of vines and rootstocks.
- Small angular yellow to brown spots on the upper leaf surface; yellow to orange powdery spore masses (pustules) on the underside; affected leaves can turn yellow and die, causing premature leaf fall (DP22).
- In severe infection the pustules can cover the entire lower leaf surface, leading to severe defoliation (NHB).

Distinguishing features

- The orange powdery pustules on the lower surface, with yellow-brown angular spots above, separate rust from downy mildew (white growth, oily spots) and powdery mildew (white-grey dust) (DP22).
- Rust is not generally reported on commercial table and wine varieties in Maharashtra; it is seen mainly on rootstocks such as Dogridge, 1103P, 110R, Ramsey and SO4 (DP22).

2. Causes and progression

- Pathogen: Phakopsora vitis (DP22, NG).
- Survival and spread: the pathogen reproduces and survives in spots on leaves or stems and in fallen host debris; secondary infection through urediniospores continually re-infects vines (DP22).
- Regional importance: NIPHM lists rust as a disease of regional significance (NG); ICAR-NRCG scientists describe rust and bacterial leaf spot as minor diseases gaining importance due to climate change (KR).

3. Favourable conditions

- Moisture and temperatures above 20 C (DP22).
- In Bangalore, weather during July-December favours the disease (NHB).

4. Prevention and management

- DP22 gives no rust-specific cultural measures. Because the pathogen survives in fallen host debris and on rootstock vines (DP22), watch rootstock and sucker growth in vineyards on these rootstocks.
- Scout leaf undersides for orange pustules in warm, wet weather; confirm with a diagnostic laboratory if unsure.

Regional note

Reported on rootstocks in Maharashtra (DP22) and on Bangalore Blue in Karnataka (NHB).

5. Remedies and treatment

- The CIB&RC list as reproduced in DP22 gives one grape rust entry: thiophanate-methyl 70% WP (V3 below). NHB (legacy) recommended 3-4 fortnightly sprays of chlorothalonil at 0.2% in July-August and January-February for rust on Bangalore Blue (NHB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiophanate-methyl 70% WP	Rust	Per label (dose not given in DP22)	-	V3	Label waiting period 7 d (DP22). Not on A5; check residue limits for export.
Chlorothalonil	Rust on Bangalore Blue	0.2%, 3-4 sprays at fortnightly intervals (July-August, January-February)	-	V2	NHB legacy publication only; chlorothalonil has grape label claims for downy mildew and anthracnose, not rust, in DP22 tables. Confirm the label.

- Thiophanate-methyl is also on A5 for anthracnose (PHI 73 d) but not for rust, so use A5 PHI values only for the anthracnose use. Check residue limits before use in export vineyards.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[KR] KRISHI (ICAR) - Sawant, I.S., Sawant, S.D. and Saha: Integrated disease management strategies in grapes (2016), abstract. https://krishi.icar.gov.in/jspui/handle/123456789/64829

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

[NG] NIPHM and DPPQS - AESA based IPM package for Grapes (pest list: diseases of national and regional significance). https://niphm.gov.in/IPMPackages/Grapes.pdf

8. Black Rot

Grape (Vitis vinifera) | Guignardia bidwellii | Fungal (warm, wet weather)

At a glance. A disease of warm, moist weather that attacks leaves, stems, tendrils, flowers and berries, ending in hard black mummified berries. Sanitation and copper-based protection of young bunches are the main measures in the sources reviewed.

1. Identification

Symptoms

- Crop / stage: leaves, stems, flowers and berries; all new growth is prone to attack during the growing season (NHB).
- Irregularly shaped reddish-brown spots on leaves and a black scab on berries; occasionally small elliptical dark cankers on young stems and tendrils (NHB).
- Affected berries shrivel and become hard black mummies (NHB).

Distinguishing features

- NIPHM lists Guignardia bidwellii as black rot of grape (NG). Do not confuse it with Aspergillus niger 'black rot' of stored berries, which is a post-harvest mould (NHB; see chapter 6).

2. Causes and progression

- Pathogen: Guignardia bidwellii (NHB, NG).
- Progression: leaf, cane and tendril infection occurs only while the tissue is young, but berries can be infected until almost fully grown if no active fungicide residue is present (NHB).

3. Favourable conditions

- A warm and moist climate with extended periods of rain and cloudy weather (NHB).

4. Prevention and management

- Collect and destroy mummified berries left on the vines (NHB).
- Ensure free air circulation through cultivation practices (NHB).
- Copper fungicides are preferred on bunches because they leave no visible deposits on the fruit surface (NHB).

Regional note

Listed among diseases of national significance in the NIPHM grape package (NG); no state-specific data found.

5. Remedies and treatment

- No black rot entry was found in the A5 label-claim list or the CIB&RC tables reproduced in DP22. The NHB legacy recommendation is one or two sprays of Bordeaux mixture (4:4:100) on young bunches (NHB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Bordeaux mixture 4:4:100	Black rot (NS)	1-2 sprays on young bunches	-	V2	NHB legacy publication only; no black rot label claim matched. Confirm any copper product's label and residue limits.

- No chemical is verified for black rot; use cultural measures first and ask ICAR-NRCG, the state horticulture department or the nearest KVK for a current recommendation.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[NG] NIPHM and DPPQS - AESA based IPM package for Grapes (pest list: diseases of national and regional significance). https://niphm.gov.in/IPMPackages/Grapes.pdf

9. Bitter Rot

Grape (Vitis vinifera) | Greeneria uvicola | Fungal (field, storage and transit losses)

At a glance. A leaf, cane and berry disease that causes losses in the field, in storage and in transit, and is most serious on older leaves. The sources reviewed give only older cultural and chemical advice, so no chemical is verified.

1. Identification

Symptoms

- Crop / stage: leaves, canes and berries; most serious on older leaves (NHB).
- Leaves: dark brown water-soaked spots covering the whole leaf lamina on either side of the veins and veinlets (NHB).
- Canes: infection first appears white and later turns black; infected canes grow less and wilt (NHB).
- Berries: young green berries shrivel, turn black and mummify; raisins made from infected berries taste sour and keep poorly (NHB).

Distinguishing features

- NIPHM lists 'Greenaria bitter rot' (Greeneria uvicola) as a disease of national significance (NG).
- Mummified berries are also a feature of black rot and downy mildew; bitter rot is separated by the water-soaked leaf lesions across veins and the white-then-black cane symptoms (NHB).

2. Causes and progression

- Pathogen: Greeneria uvicola (NHB, NG).
- Detailed survival and spread information was not found in the sources reviewed.

3. Favourable conditions

- Specific weather thresholds were not found in the sources reviewed.

4. Prevention and management

- NHB (citing IIHR, Bangalore) recommends pruning of canes as the first step, followed by protective sprays (NHB).

Regional note

No state-specific data found.

5. Remedies and treatment

- The NHB legacy list names iprodione, bitertanol and thiophanate-methyl for field and storage control. None of these has a bitter rot label claim in A5 or in the CIB&RC tables reproduced in DP22, so none is recommended.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
No product verified	Bitter rot	-	-	--	No bitter rot entry in A5 or in the DP22 CIB&RC tables; NHB legacy products not reproduced.

- Ask ICAR-NRCG, the state horticulture department or the nearest KVK for a current recommendation.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[NG] NIPHM and DPPQS - AESA based IPM package for Grapes (pest list: diseases of national and regional significance). https://niphm.gov.in/IPMPackages/Grapes.pdf

10. Dead Arm (Phomopsis Cane and Leaf Spot)

Grape (Vitis vinifera) | Phomopsis viticola | Fungal (cane- and wood-borne)

At a glance. A disease mostly reported in South India that causes angular leaf spots and brown cane lesions, and in severe cases drying of woody parts, roots and the whole vine. Pruning out affected wood and protecting fresh cuts are the sourced measures.

1. Identification

Symptoms

- Crop / stage: leaves, stems, canes and flower clusters (NHB).
- Small angular spots on leaves, stems, canes and flower clusters, mostly with yellowish margins and dark centres; spots often join to form large brown areas on canes, and the canes then dry rapidly (NHB).
- In severe cases the fungus spreads into the woody part and attacks water-conducting tissues; the drying can extend to the roots and the whole plant wilts (NHB).

Distinguishing features

- NIPHM lists Phomopsis viticola leaf spot as a disease of regional significance, reported from Andhra Pradesh (NG).
- Cane drying also arises from Botryodiplodia (irregular cane maturity, NG) and from anthracnose cankers (DP22); confirm the cause with a laboratory.

2. Causes and progression

- Pathogen: Phomopsis viticola (NHB, NG).
- Progression: from leaf and cane spots into the woody tissue, then roots, in severe cases (NHB). The disease is mostly prevalent in South India (NHB).

3. Favourable conditions

- Specific weather thresholds were not found in the sources reviewed.

4. Prevention and management

- Collect and destroy pruned canes (NHB).
- Prune dead canes back to healthy tissue, and further if needed, so no mycelium is left in the canes; paste pruned cuts with Bordeaux paste immediately after pruning (NHB). The ICAR-NRCG bulletin also advises removing dead wood and pasting cut surfaces with Bordeaux paste, or spraying 1% Bordeaux mixture, within 1-2 days of pruning (TB8).

Regional note

Mostly South India (NHB); reported from Andhra Pradesh (NG).

5. Remedies and treatment

- NHB (legacy) follows pruning with Bordeaux mixture (5:5:50) and then older fungicides at fortnightly intervals until the canes harden. The older products named there are not on A5 and are not reproduced.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Bordeaux paste on pruning cuts; Bordeaux mixture 5:5:50 spray	Dead arm	Paste immediately after pruning; spray afterwards	-	V2	NHB and TB8 (legacy). No dead-arm label claim matched. Confirm the label for any copper product.

- No chemical is verified for dead arm. Pruning hygiene is the main sourced control.

6. Sources and citations

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[TB8] ICAR-National Research Centre for Grapes - Technical Bulletin No. 8: Package of practices for managing major diseases and insect pests on grapes (March 2007). https://nrcgrapes.icar.gov.in/sites/default/files/publications/techbul-8-pop-for-disease-insect-pest-on-grapes.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

[NG] NIPHM and DPPQS - AESA based IPM package for Grapes (pest list: diseases of national and regional significance). https://niphm.gov.in/IPMPackages/Grapes.pdf

11. Esca (Black Measles)

Grape (Vitis vinifera) | Phaeomoniella chlamydospora, Phaeoacremonium minimum (syn. P. aleophilum) and other Phaeoacremonium spp.; Fomitiporia in late wood decay | Fungal (wood-infecting trunk-disease complex)

At a glance. A trunk-disease complex that shows as striped, scorched leaves and dark-spotted berries on mature vines, with dark streaks inside the wood. The fungi enter through pruning wounds and cannot be removed by fungicides once established, so wound protection and sanitation are the measures. Esca is not described in the Indian sources used for the other chapters of this document (NHB, TB8, DP22, NIPHM). This chapter rests on UC IPM and other international research and carries no verified Indian chemical.

1. Identification

Symptoms

- Crop / stage: leaves, berries and wood of established vines. Symptoms usually become apparent in vineyards 5 to 7 or more years old, although the infections occur in younger vines (UE).
- Leaves: interveinal 'striping'. The stripes start dark red in red cultivars and yellow in white cultivars, then dry and turn necrotic. They can appear at any time in the season but are most common in July and August, are often limited to one shoot or to shoots from the same spur or cane, and affected leaves can dry completely and drop early (UE).
- Berries: small, round dark spots, each bordered by a brown-purple ring, easiest to see on white cultivars and appearing any time from fruit set to ripening; in severely affected vines berries crack and dry or spoil (UE). The spots can run together and make berries look black (GM).
- Wood: cross-sections of canes, spurs, cordons or trunks under symptomatic shoots show concentric rings of dark spots (UE). Cut sections of arms, cordons or trunks often ooze a dark sap, and older vines also carry secondary wood-rotting fungi (GM).
- Symptoms are inconsistent from year to year, which UC attributes to possibly climate-linked variation in toxin production. An acute form, apoplexy, kills a shoot or adjacent shoots suddenly and is more common in Europe (UE).

Distinguishing features

- Esca is one of four trunk diseases (with Botryosphaeria, Eutypa and Phomopsis dieback). Dead spurs and shoot dieback are shared symptoms, and mixed infections occur even within one vine (UEU). Leaf striping together with ringed berry spots on the same shoot points to esca (UE).
- Wedge-shaped wood cankers from Eutypa, Botryosphaeria and Phomopsis dieback cannot be told apart by eye (UEU). Phomopsis dead arm (chapter 10) and anthracnose cankers (chapter 3) also dry canes; confirm the cause with a laboratory.
- Black rot (chapter 8) gives reddish-brown leaf spots and shrivelled, hard black mummified berries (NHB), not ringed spots on berries of shoots with striped leaves.

2. Causes and progression

- Pathogens: a complex that includes Phaeoacremonium aleophilum (now named P. minimum; sexual stage Togninia minima), other Phaeoacremonium species and Phaeomoniella chlamydospora (UE, TRI). The wood-rotting basidiomycete Fomitiporia mediterranea is also part of the esca complex (SU).
- Survival: the structures that produce spores (perithecia or pycnidia, depending on the fungus) are embedded in diseased woody parts of vines (UE).
- Infection: spores are released during fall-to-spring rainfall, and wounds made by dormant pruning are the infection sites. Wounds can stay susceptible for several weeks, with susceptibility declining over time (UE). In UC trials, inoculating pruning wounds with P. chlamydospora or P. aleophilum reproduced typical esca symptoms (UCT).
- Once a pruning wound is infected, the pathogen establishes a permanent, localized wood infection that cannot be eradicated by fungicide applications (UE).
- Nursery link: in a nursery study most bench-grafted and one-year-old vines showed wood blackening and harboured Phaeoacremonium spp. and P. chlamydospora, whereas cuttings showed very little (ZA). In young vineyards the same fungi are linked to Petri disease and decline (GTD).
- Losses: UC notes that the economic impact on grape production is difficult to measure accurately (OC4H). No Indian loss data were found.

3. Favourable conditions

- Rain in the dormant season releases spores, and all wounds made in that season are susceptible, including pruning cuts made to reposition spurs (UE, UEU).
- Expression of leaf and fruit symptoms varies with the season and is thought to follow climate-related toxin production (UE).
- The sources reviewed give no weather thresholds for Indian vineyards, and they do not show whether India's April and October pruning cycles coincide with spore release.

4. Prevention and management

- Preventive practices (delayed pruning, double pruning and pruning-wound protectants) are the most effective approach for all trunk diseases. They work best when started in young vineyards (under 5 years) and used every year (UEU).
- In California, delaying pruning as late in the dormant season as possible (February or later) lowers risk by reducing wound susceptibility and avoiding the peak spore release of December and January rains (UEU). This is a California rule; the sources reviewed do not test it for India's pruning calendar.
- Double pruning, for large cordon-trained, spur-pruned vineyards: pre-prune early in winter, cutting canes 12 to 18 inches above the final cuts, then hand-prune to spurs later (UEU).
- In diseased mature vineyards, sanitation and vine surgery are less effective and costlier than prevention. In summer, when rain is unlikely, cut cankered parts back to healthy wood, remove and destroy the woody debris, and retrain a new trunk or cordon where needed (UEU).
- Use healthy planting material: esca-associated fungi were found in most bench-grafted vines in a nursery study (ZA).
- A wound sealant of 5% boric acid in acrylic paint (Tech-Gro B-Lock) is reported effective against Eutypa dieback and esca, and an essential-oil sealant (Safecoat VitiSeal) is also listed (UEU). These are US products.
- India: chapter 10 gives Bordeaux paste on pruning cuts for dead arm (NHB, TB8); no source reviewed tests it against esca.

Regional note

No Maharashtra, Karnataka, Andhra Pradesh or Tamil Nadu recommendation for esca was found. The measures above come from UC IPM and other international sources.

5. Remedies and treatment

- There is no curative treatment. UC states that infection cannot be eradicated by fungicide applications once a wound is infected (UE).
- UC IPM lists pruning-wound protectants for trunk diseases: thiophanate-methyl (paint or spray; a US Special Local Needs label), myclobutanil and tetraconazole (UEU). US label doses are not reproduced here.
- In the tables of this document (A5, DP22), thiophanate-methyl is registered on grape for anthracnose (chapter 3) and myclobutanil and tetraconazole for powdery mildew (chapter 2). None is listed for esca or pruning-wound protection. A5 also notes that resistance to triazoles has been detected in powdery mildew in India.
- A Trichoderma atroviride SC1 product (Vintec) is used against grapevine trunk diseases in Europe and reduces wood colonization (TRI). Its registration in India was not verified.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiophanate-methyl	Pruning-wound protection (esca, Eutypa), US	See US label	-	--	UC IPM (UEU). In India A5 lists it for anthracnose only; no esca claim. Not recommended here.
Myclobutanil; tetraconazole	Pruning-wound protection, US	US doses not reproduced	-	--	UC IPM (UEU). In India A5 lists them for powdery mildew only (triazole resistance noted).
5% boric acid in acrylic paint; essential-oil sealant	Wound sealant, US products	Per product	-	--	UC IPM (UEU). Indian registration not assessed.
Trichoderma atroviride SC1	Biological wood protection, Europe	Per product	-	--	TRI. Indian registration not verified.

- No chemical is verified for esca in India and none is recommended. Ask ICAR-NRCG, the state horticulture department or the nearest KVK for a current recommendation.
- Tags follow this document's scheme: '--' means no chemical option verified for this disease; none is recommended.
- A5 and DP22 are the sources of this document's earlier tables; the CIB&RC register itself was not searched again for esca or pruning-wound claims.

6. Sources and citations

[UE] UC IPM Pest Management Guidelines: Grape - Esca (Black Measles). UC ANR Publication 3448 (text updated 12/14). https://ipm.ucanr.edu/agriculture/grape/esca-black-measles/

[UEU] UC IPM Pest Management Guidelines: Grape - Eutypa Dieback (text updated 12/14; treatment table updated 07/15); the Esca page refers to it for management. https://ipm.ucanr.edu/agriculture/grape/eutypa-dieback/

[GM] eXtension Grapes - Grapevine measles (esca, black measles; summary of UC material). https://grapes.extension.org/?p=708

[UCT] UC research report on inoculating grape pruning wounds with Phaeomoniella chlamydospora and Phaeoacremonium aleophilum (esca symptoms reproduced). https://class.ucanr.edu/sites/default/files/2019-05/303239.pdf

[OC4H] UC Cooperative Extension Fresno County: grapevine canker diseases and pruning-wound protection report. https://oc4h.ucanr.edu/sites/default/files/2012-03/139208.pdf

[SU] Surico, G., Mugnai, L. & Marchi, G. (2006). Older and more recent observations on esca: a critical overview. Phytopathologia Mediterranea 45(Suppl.): 68-86 (abstract). https://oajournals.fupress.net/index.php/pm/article/view/5190

[ZA] Zanzotto, A., Borgo, M., Serra, S. & Viel, W. (2001). Investigation into the occurrence of esca-associated fungi in cuttings and bench-grafted vines. Phytopathologia Mediterranea 40(4): 311-316 (abstract). https://oajournals.fupress.net/index.php/pm/article/view/4916

[TRI] Comparative transcriptomics suggests early modifications by Vintec in grapevine trunk in response to Phaeomoniella chlamydospora and Phaeoacremonium minimum (PMC9152730). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9152730/

[GTD] Diaz, G.A., Reveglia, P., Tomoiaga, L.L. & Chedea, V.S. (2023). Editorial: Fungal pathogens causing the grapevine trunk diseases - biology and identification. Frontiers in Fungal Biology (PMC10512359). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10512359/

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes, April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

TB8 and NIPHM (NG, NP) are cited in this chapter by their code from the sources listed earlier in this document.

Consolidated source list and items not included

[A5] ICAR-National Research Centre for Grapes (Pune) - Annexure-5: List of chemicals with CIB&RC label claim for use in grapes (revised 29 October 2024). Hosted by APEDA. https://apeda.gov.in/apedahindi/Grapenet/Annexure_5.pdf

[TB8] ICAR-National Research Centre for Grapes - Technical Bulletin No. 8: Package of practices for managing major diseases and insect pests on grapes (March 2007). https://nrcgrapes.icar.gov.in/sites/default/files/publications/techbul-8-pop-for-disease-insect-pest-on-grapes.pdf

[NHB] National Horticulture Board, Government of India - Grapes: Diseases. https://www.nhb.gov.in/pdf/fruits/grape/gra002.pdf

[NP] NIPHM and Directorate of Plant Protection, Quarantine & Storage - Grapes pests list (via Vikaspedia). https://en.vikaspedia.in/viewcontent/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-pests?lgn=en

[VK] Vikaspedia (IPM strategies for grapes) - Grapes: diseases and symptoms. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-grapes/grapes-diseases-and-symptoms

[KR] KRISHI (ICAR) - Sawant, I.S., Sawant, S.D. and Saha: Integrated disease management strategies in grapes (2016), abstract. https://krishi.icar.gov.in/jspui/handle/123456789/64829

[MDM] Michigan State University IPM - Downy mildew (grape). https://www.canr.msu.edu/ipm/diseases/downy_mildew

[UDM] UC IPM - Downy mildew of grape (home and landscape). https://ipm.ucanr.edu/home-and-landscape/downy-mildew-of-grape/

[VT] Vineyard Team (UC IPM-based) - Downy mildew (Plasmopara viticola): damage, symptoms and life cycle. https://www.vineyardteam.org/resources/resource-library/pest-management.php?id=611

[CIB] CIB&RC, DPPQS - Major Uses of Pesticides: Fungicides, up to 30/09/2025 (hosted copy used for spot cross-checks). https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[DP22] Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India - IPM Package of Practices for Grapes (for producing quality fruits for export), April 2022; reviewed by ICAR-National Research Centre for Grapes, Pune. Includes CIB&RC waiting-period tables compiled from Major Uses of Pesticides (01/01/2021). https://ppqs.gov.in/sites/default/files/ipm_pop_grapes.pdf

[NG] NIPHM and DPPQS - AESA based IPM package for Grapes (pest list: diseases of national and regional significance). https://niphm.gov.in/IPMPackages/Grapes.pdf

Items checked but not included

- Foot rot (Cylindrocarpon spp.), irregular cane maturity due to Botryodiplodia, and angular leaf spot (listed as a mancozeb label claim in DP22): named in NIPHM / DP22 lists but with no description to support a chapter.
- Streptocycline (NHB, bacterial leaf spot) and older products named in NHB (benomyl, Bayleton, Calaxin, Baycor, Rovral, Difolatan, Daconil): not on the A5 or DP22 lists, so not reproduced.
- Doses for V3 products: not available in the sources used; see the label.
- Insect pests, nematodes, weeds, plant growth regulators and herbicides in A5, TB8 and DP22: outside the scope of a disease document.
- Viral diseases of grape (for example leaf roll): not covered in the sources reviewed.
- Registration changes after 29 October 2024 (A5), after 01/01/2021 (label waiting periods in DP22) and after 30/09/2025 (CIB&RC compilation): not checked.
- Bio-fungicide registrations (Trichoderma, Bacillus subtilis): not verified against the CIB&RC bio-pesticide list, so they are tagged V2.

Consolidated source list (continued): sources added with chapter 11

[UE] UC IPM Pest Management Guidelines: Grape - Esca (Black Measles). UC ANR Publication 3448 (text updated 12/14). https://ipm.ucanr.edu/agriculture/grape/esca-black-measles/

[UEU] UC IPM Pest Management Guidelines: Grape - Eutypa Dieback (text updated 12/14; treatment table updated 07/15); the Esca page refers to it for management. https://ipm.ucanr.edu/agriculture/grape/eutypa-dieback/

[GM] eXtension Grapes - Grapevine measles (esca, black measles; summary of UC material). https://grapes.extension.org/?p=708

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[OC4H] UC Cooperative Extension Fresno County: grapevine canker diseases and pruning-wound protection report. https://oc4h.ucanr.edu/sites/default/files/2012-03/139208.pdf

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[TRI] Comparative transcriptomics suggests early modifications by Vintec in grapevine trunk in response to Phaeomoniella chlamydospora and Phaeoacremonium minimum (PMC9152730). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9152730/

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Items checked but not included (chapter 11 addition)

- Indian occurrence records for esca: searched for and not found in ICAR-NRCG, NHB, DPPQS or NIPHM material; no Indian loss figures.
- CIB&RC register entries for pruning-wound protection or esca on grape: not searched again; the A5 and DP22 tables used in this document list none.
- US label doses for wound protectants and brand-name sealants: not reproduced.
- Other trunk diseases (Botryosphaeria dieback, Eutypa dieback, Petri disease and black-foot of young vines): not covered as chapters.
- Experimental approaches (essential oils, Trichoderma in Europe, toxin and cavitation studies): mentioned only where cited.